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Course –ECA5601- 5g/6g Communication

Experiment: 9

Massive MIMO Capacity vs SNR and Gain Using MATLAB

Aim

To analyze the performance of a Massive MIMO system by studying channel capacity, signal-to-noise ratio (SNR), and antenna gain using MATLAB.

Objectives

Objective – 1: Analyze Channel Capacity (bps/Hz)

Analyze the channel capacity of a Massive MIMO system for different signal-to-noise ratio (SNR) values using MATLAB. Observe how channel capacity increases with improving signal quality based on the Shannon capacity principle.

Objective – 2: Compare Signal-to-Noise Ratio (SNR) (dB)

Compare the signal-to-noise ratio for different communication scenarios using MATLAB. Analyze the effect of SNR on the quality and reliability of wireless communication in Massive MIMO systems.

Objective – 3: Analyze Antenna Gain (dB)

Analyze the antenna gain achieved by varying the number of antenna elements in a Massive MIMO system using MATLAB. Observe how increasing the antenna array size improves directional gain and enhances communication performance.

Procedure

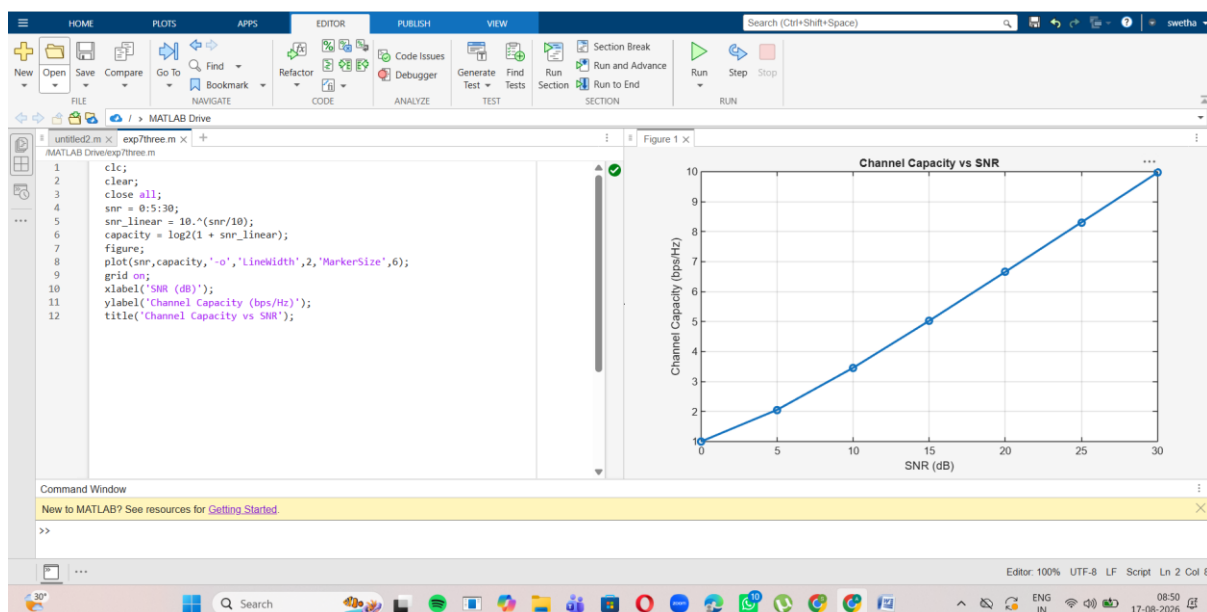
1. Define the number of transmit and receive antennas, SNR range, and different channel gain values in MATLAB.
2. Generate a random MIMO channel matrix to represent the communication link between the multiple transmit and receive antennas.
3. Calculate the channel capacity for each SNR value while considering the specified channel gains.
4. Store the calculated capacity values and repeat the process for all selected gain values to enable comparison.
5. Plot the capacity against SNR for different gain values and analyze how increasing SNR and channel gain affects the overall Massive MIMO capacity.

objective 1 :

Program

```
clc;
clear;
close all;
snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);
figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');
```

output



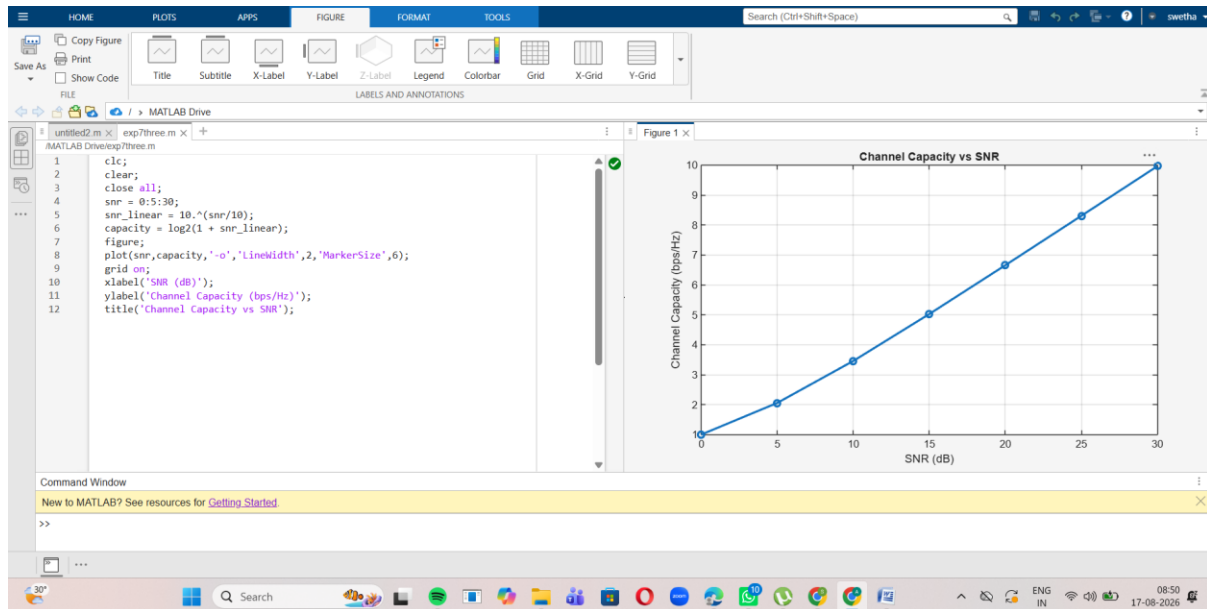
Objective 2: code and output:

```
clc;
clear;
close all;
snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);
figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
grid on;
xlabel('SNR (dB)');
```

```

ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');
% Array Gain (dB)
xlabel('Steering Angle (Degrees)')
ylabel('Array Gain (dB)')
title('Array Gain for Different Steering Angles')

```

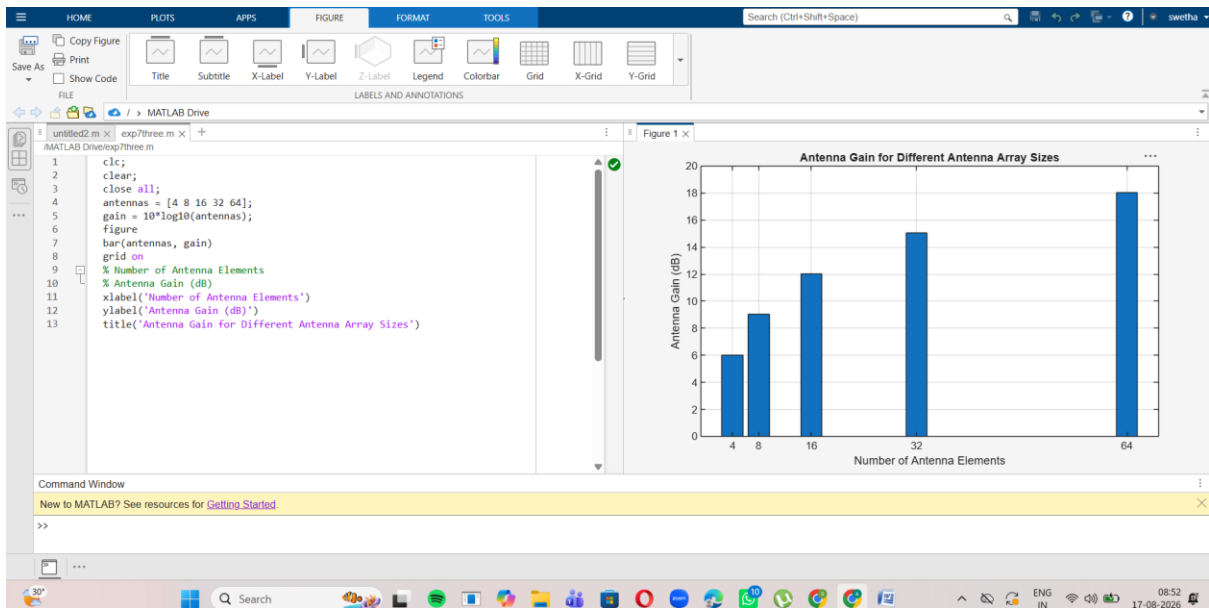


Objective 3:code and output:

```

clc;
clear;
close all;
antennas = [4 8 16 32 64];
gain = 10*log10(antennas);
figure
bar(antennas, gain)
grid on
% Number of Antenna Elements
% Antenna Gain (dB)
xlabel('Number of Antenna Elements')
ylabel('Antenna Gain (dB)')
title('Antenna Gain for Different Antenna Array Sizes')

```



Result:

The MATLAB analysis successfully demonstrated the relationship between channel capacity, signal-to-noise ratio, and antenna gain in a Massive MIMO system. The results showed that higher SNR and larger antenna arrays improve communication performance by increasing channel capacity and directional gain.